Week 6 BA

2024 Aug04 Quiz-2

Comprehension for questions 1-5 (source Q23)

A multiple linear regression model, as specified below is fit on a data set with 50 data points. Then answer the given sub questions

$$\text{MLR Model: } Y = 4.1 - 0.4 * X_1 + 4.2 * X_2 - 0.5 * X_3 + \varepsilon$$

1. [Source Q24; SA]
How many degrees of freedom are present for the "Residuals" in the ANOVA Table?

[Source ans: 46]

2. [Source Q25; SA]
How many degrees of freedom are present for the "Regression" in the ANOVA Table?

[Source ans: 3]

3. [Source Q26; SA]
How many total degrees of freedom are present for the fitted model in the ANOVA Table?

[Source ans: 49]

4. [Source Q27; SA]
If the SSE and SST for the model are 900 and 1400 respectively, then what is the value for the model's F-statistic? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 8.4 to 8.8]

5. [Source Q28; SA]
If the SSE and SST for the model are 900 and 1400 respectively, then what is the Adjusted R-square value for model? (Note: Enter the answer in "%" rounded to two decimal places without the "%" symbol. For example, if the answer is "1.234%", then enter it as "1.23")

[Source ans: 31 to 32]

2024 Dec01 Quiz-2

6. [Source Q6; MSQ]
What does the term "Multicollinearity" refer to? (Select all that are applicable)

(a) The dependent and independent variables are not-related

(b) The dependent and independent variables are linearly related

(c) The dependent variable is linearly related to another dependent variable

(d) None of these

[Source ans: d]

Comprehension for questions 7-9 (source Q8)

A multiple linear regression model, as specified below, is fit on a dataset with 320 data points. Then answer the given subquestions *(Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")*

MLR Model: $Y = 2.1 + 1.4 * X_1 - 4.2 * X_2 + 0.5 * X_3 + 7 * X_4 - 6 * X_5 + \varepsilon$

7. [Source Q9; SA]
How many degrees of freedom are present for the "Residuals" in the ANOVA Table?

[Source ans: 314]

8. [Source Q10; SA]
How many total degrees of freedom are present for the fitted model in the ANOVA Table?

[Source ans: 319]

9. [Source Q11; SA]
If no feature engineering was performed, then how many features were present in the dataset?

[Source ans: 5]

Comprehension for questions 10-11 (source Q16)

You are solving a regression problem with 8 explanatory variables. The data has 150 observations, and the R-square value was found to be 0.75. You are adding one more explanatory variable to the dataset (a total of 9 explanatory variables). The new R-square value is 0.8. Then answer the given subquestions.

10. [Source Q17; SA]
What is the new adjusted R-square value?(Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 0.77 to 0.80]

11. [Source Q18; MCQ]
What does the value of the new adjusted R-square value imply?

(a) The new variable must be dropped from the model

(b) The new variable must be kept in the model

(c) The data is too small for fitting a regression model with 9 variables

(d) None of these

[Source ans: b]

2024 Dec22 ET

Comprehension for question 12 (source Q14)

A doctor is of the opinion that "Blood Pressure (BP)" is affected by "Age" and "Weight". Based on past data, the correlation between these three variables is identified and provided in Table-3. Then answer the given subquestions.

	BP	Age	Weight
BP	--		
Age	0.695	--	
Weight	0.950	0.507	--

Table-3

12. [Source Q16; SA]
What is the VIF for "Weight" with respect to "Age"? (Note: Enter your answer in decimal rounded to two decimal places. Eg: If your answer is "10.256", then enter it as "10.26")

[Source ans: 1.30 to 1.40]

2025 Mar16 Quiz-2

Comprehension for questions 13-18 (source Q20)

The three figures below (Picture-1, Picture-2, Picture-3) provide the partial regression output from excel for three models (Model-1, Model-2 and Model-3) which were built on the same data (Hint: same dependent variable with same number of rows. However, the number of independent variable in each model is different). Given this information, answer the subquestions.

ANOVA					
	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	**A1**	59574.54	3177.885	5.44128E-10
Residual	6	112.4796	18.7466		
Total	9	178836.1			

Picture-1: Partial Results from Excel for Model-1

ANOVA					
	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	178723.4403		5552.402929	6.27476E-12
Residual		112.6596987			
Total	9				
	Coefficients	*Standard Error*	*t Stat*		
Intercept	815.7547951	4.075009576			
X Variable 1	-3.952958885	0.03846985	**A2**		
X Variable 2	3.664290745	0.791804173			

Picture-2: Partial Results from Excel for Model-2

SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.998720528				
R Square	0.997442694				
Adjusted R Square	0.99712303				
Standard Error	7.560908499				
Observations	10				
ANOVA					
	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	**A3**		3120.29158	1.17067E-11
Residual	8	**A4**			
Total	9	**A5**			

Picture-3: Partial Results from Excel for Model-3

13. [Source Q21; SA]
What is the value of "A1"? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 178720 to 178726]

14. [Source Q22; SA]
What is the value of "A2"? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: -103.00 to -101.00]

15. [Source Q23; SA]
What is the value of "A3"? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 178373.00 to 178383.00]

16. [Source Q24; SA]
What is the value of "A4"? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 455.00 to 459.00]

17. [Source Q25; SA]
What is the value of "A5"? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 178831.00 to 178841.00]

18. [Source Q26; MCQ]
Which model is the best model for the provided data?

(a) Model-1

(b) Model-2

(c) Model-3

(d) Dr. Milo did not teach this, so none of the models are good (This is not thecorrect answer, please do not choose it. Just relax, think and answer the question)

[Source ans: b]

2025 Apr13 ET

Comprehension for question 19 (source Q16)

(The following is a purely imaginary scenario) A survey was conducted among 100 students who participated in all events at IITMs recent "Saarang" event. The students were asked to rate various aspects of their experience on a scale of "-5 to +5", where "-5" indicates a very negative experience and "+5" indicates a very positive experience. The average student rating for each event was taken across four key parameters: "Event Ambience", "Fairness of Event Judges", "Event Conduct", "Event Prize Money". The target variable was "Event Performance" which measures the overall performance of each event. Based on the collected data, a correlation matrix as specified in Table-2 is obtained for the independent variables. With this information, answer the given sub-questions

	Pairwise Correlation Values			
	Event Ambience	Fairness of Event Judges	Event Conduct	Event Prize Money
Event Ambience	1	-0.04	0.49	0.55
Fairness of Event Judges	-0.04	1	-0.011	0.022
Event Conduct	0.49	-0.011	1	-0.51
Event Prize Money	0.55	0.022	-0.51	1

Table-2

19. [Source Q19; SA]
What will be the percentage increase in the standard error corresponding to the beta value of "Event Ambience", if a multiple linear regression model where both "Event Ambience" and "Event Price Money" are used as explanatory variables to predict "Event Performance"? (Note: Enter the answer in percentage rounded to two decimal places. For example, if the answer is "1.234%", then enter it as "1.23")

[Source ans: 18.00 to 20.00]

2025 Aug03 Quiz-2

Comprehension for questions 20-26 (source Q5)

Milo's Bakery is famous for three products "BA Biscuits (BAB)", "TA Controversy Cake (TACC)" and "Hot Argument Drink (HAD)". Dr.Milo, the owner of the shop, is an aspiring data analyst. He has collected 5 days of data. On each day, he captures the number of each product made and total kilogram of cooking gas consumed. Using all the collected data, Dr.Milo has built a linear regression model in excel, and the partial regression output is provided in Figure-2. Using this information, answer the given subquestions.

Regression Statistics				
Multiple R				
R Square				
Adjusted R Square	0.97			
Standard Error	3.36			
Observations				
ANOVA				
	df	*SS*	*MS*	*F*
Regression		Q2		Q5
Residual	Q1	Q3		
Total		Q4		
	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	23.8	13.2		0.32
No. of BAB made	0.18	0.05		0.17
No. of TACC made	0.32	0.16		0.3
No. of HAD made	-0.008	0.1		0.95

Figure-2: Partial Regression Output from Excel

20. [Source Q6; SA]
What is the sample size for building the model in Figure-2?

[Source ans: 5]

21. [Source Q7; SA]
What is the value of Q1?

[Source ans: 1]

22. [Source Q8; SA]
What is the value of Q2? (Note: Enter your answer rounded to two decimal places. For example, if your answer is "1.235" then enter it as "1.24")

[Source ans: 1490.00 to 1496.00]

23. [Source Q9; SA]
What is the value of Q3? (Note: Enter your answer rounded to two decimal places. For example, if your answer is "1.235" then enter it as "1.24")

[Source ans: 11.10 to 11.40]

24. [Source Q10; SA]
What is the value of Q4? (Note: Enter your answer rounded to two decimal places. For example, if your answer is "1.235" then enter it as "1.24")

[Source ans: 1503.00 to 1508.00]

25. [Source Q11; SA]
What is the value of Q5? (Note: Enter your answer rounded to two decimal places. For example, if your answer is "1.235" then enter it as "1.24")

[Source ans: 43.00 to 45.00]

26. [Source Q12; MCQ]
At a 15% significance level, Dr. Milo thinks that the gas consumption at his company is affected by the number of BA, TACC and HAD that are made in a day. Then what do you conclude about Dr. Milo?

(a) Dr. Milo is dumb and does not know statistics.

(b) Dr. Milo is a genius and he is right.

(c) Cannot comment on Dr. Milo's conclusion without more information

[Source ans: a]

2025 Aug31 ET

27. [Source Q2; MSQ]

Which of the following Regression equations is/are MLR? (Choose all that is/are applicable)

(a) $Ln(Y) = Beta_0 - Beta_1 * X_1 + Beta_2 * X_2$

(b) $Ln(Y) = Beta_0 + Beta_1 * X_1 + Beta_2 * X_2$

(c) $Y = Beta_0 - Beta_1 * X_1 + Beta_2 * X_2 + Beta_3 * X_1 * X_2$

(d) $Y = Beta_0 - Beta_1 * X_1 + Beta_2 * X_2 + Beta_3 * (X_1)^2 * SQRT(X_2)$

(e) $Ln(Y) = Beta_0 - Beta_1 * X_1 + Beta_2 * X_2 + (Beta_3)^2 * X_1 + Beta_4 * Ln(X_2)$

(f) None of these are MLR models

[Source ans: a, b, c, d]

Comprehension for question 28 (source Q13)

The Following Comprehension and Related Subquestions are Purely Hypothetical Table-2 provides the Cumulative Television Rating Points (CTRP), money spend on promotion (in Lakhs, denoted by "P"), and the advertisement revenue (in Lakhs, denoted by "R") generated over a one-month period for 8 different television programs. The Revenue a TV show makes is the Dependent Variable. Using this data, answer the given subquestions.

TV Program ID	CTRP	P (in Lakhs)	R (in Lakhs)
TVP001	133	1.11	11.97
TVP002	111	1.04	10.53
TVP003	129	0.97	11.24
TVP004	117	0.79	9.87
TVP005	130	0.13	12.83
TVP006	154	1.08	12.95
TVP007	149	1.47	14.07
TVP008	90	1.04	9.22

Table-2

28. [Source Q16; SA]

An MLR model to predict "R" is built using all the data in Table-2. If the built model explains ~82.3% of variability in "R", then what is the value for the Model's F-statistic?

[Note: Enter your answer in decimal rounded to two decimal places. For example, if your answer is 1.235 then enter the answer as "1.24"]

[Source ans: 11.00 to 12.00]

Shiva(25f1000035)

2026 Apr06 Quiz-2

Comprehension for questions 29-31 (source Q1)

Milo Metals, a metal works shop, wants to predict the "Profit" for its products based on the expenditure spent on areas like "Quality", "Design" and "Delivery Time". Accordingly, data was collected for 50 different products it manufactured and sold. Milo Jr., the great grand son of Milo Sr., having joined the BS program, is interested to use regression as a tool to predict the profit. Accordingly, using the 50 sample points, he build the following FOUR models Model-1: Profit vs Quality Model-2: Profit vs Design Model-3: Profit vs Delivery Time Model-4: Profit vs Quality,Design,Delivery Time The partial regression outputs for the FOUR models are provided in Figures 1 to 4. Additionally, Table-1 provides the correlation matrix for the various columns in the data set.

Regression Statistics	
Multiple R	
R Square	
Adjusted R Square	0.95
Standard Error	9.4164
Observations	

Figure-1: Partial Regression Output for Model-1

Using this information, answer the given sub-questions

ANOVA					
	df	*SS*	*MS*	*F*	*Significance F*
Regression		**P1**		**P2**	
Residual		76397.77			
Total					

Figure-2: Partial Regression Output for Model-2

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	90.07	8.239		1.26E-14
Delivery Time	0.25	0.032	**P3**	4.38E-10

Figure-3: Partial Regression Output for Model-3

ANOVA				
	df	*SS*	*MS*	*F*
Regression	**P4**	75683.9642		
Residual	**P5**			
Total				

Figure-4: Partial Regression Output for Model-4

	Profit	*Quality*	*Design*	*Delivery Time*
Profit	1			
Quality	0.97	1		
Design	0.2	0.24	1	
Delivery Time	0.75	0.72	-0.03	1

Table-1: Correlation Matrix

29. [Source Q7; SA]
What is the value for P4? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 3]

30. [Source Q8; SA]
What is the value for P5? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 46]

31. [Source Q9; SA]
What is the VIF between "Design" and "Delivery Time" when a regression model to predict "Price" is built using these two variables? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 1.10 to 1.20]

2026 May06 ET

32. [Source Q28; MSQ]
What does the term "Multicollinearity" refer to? (Select all that are applicable)

(a) The dependent and independent variables are not-related

(b) The dependent and independent variables are linearly related

(c) The dependent variable is linearly related to another dependent variable

(d) The dependent variables are related among each other

(e) None of these

[Source ans: d]